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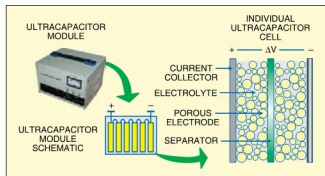


Fig. 2: Schematic of an ultracapacitor module

Advantages of ultracapacitors

1. High number of charge-discharge cycles (millions or more compared to 200-1000 for most commercially available rechargeable batteries)
2. No disposable parts during the entire operating life of the device, which makes them environment-friendly
3. Extremely low internal resistance, high efficiency (97-98 per cent), high output power, extremely low heating levels and improved safety

an electrochemical device, no chemical reaction is involved in its energy storage mechanism. This mechanism is highly reversible, and allows the ultracapacitor to charge and discharge hundreds of thousands of times.

Once the ultracapacitor is charged and energy is stored, the load can use this energy. The amount of energy stored is very large compared to a standard capacitor because of the enormous surface area created by the porous carbon electrodes and the small charge separation created by the dielectric separator.

Technical description. An ultracapacitor can be viewed as two non-reactive porous plates, or collectors, suspended within an electrolyte, with a voltage potential applied across the collectors. In an individual ultracapacitor cell, the applied potential on the positive electrode attracts the negative ions in the electrolyte, while the potential on the negative electrode attracts the positive ions. A dielectric separator be-

tween the two electrodes prevents the charge from moving between the two electrodes.

Electrical energy storage devices, such as capacitors, store electrical charge on an electrode. Electrochemical cells or batteries utilise the

electrode to create, by chemical reaction, an electrical charge at the electrodes. In both the cases the ability to store or create electrical charge is a function of the surface area of the electrode. Ultracapacitor stores energy by relying on the microscopic charge separation at an electrochemical interface.

In contrast with traditional capacitors, ultracapacitors don't have a conventional dielectric. These capacitors use 'plates,' which are, in fact, two layers of the same substrate, the so-called 'electrical double layer,' resulting in the effective separation of charge despite the vanishingly thin physical separation of the layers. Absence of a bulky layer of dielectric permits the packing of 'plates' with much larger surface area into a given size, resulting in their extraordinarily high capacitances in practical sized packages.

However, ultracapacitors can withstand only low voltage, which means ultracapacitors rated for higher voltages must be made of matched series-connected individual electric double-layer capacitors, much like series-connected cells in higher-voltage batteries. The capacitors improve storage density through the use of a non-porous material, typically activated charcoal, in place of the conventional insulating barrier.

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